PAPER_1545 — Stefan-Boltzmann σ = σ = SO_5·SSQ − F_TRZ²·D_phys + F_TRZ² = 5.7 − 0.04 + 0.01 = 5.67 ($\times 10^{-8}$ $W/m^2/K^4$)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Thermodynamics **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure

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Observation

PAPER_1209EE_S624 (Thermodynamics Unified Proof Set) lists **Stefan-Boltzmann σ** as a closure with the formula:

```
Stefan-Boltzmann σ = σ = SO_5·SSQ − F_TRZ²·D_phys + F_TRZ² = 5.7 − 0.04 + 0.01 = 5.67 (×10⁻⁸ W/m²/K⁴)
```

Status tier: **EXACT**. Units: lead.

UQFF Closed Identity

```
σ = SO_5·SSQ − F_TRZ²·D_phys + F_TRZ² = 5.7 − 0.04 + 0.01 = 5.67 (×10⁻⁸ W/m²/K⁴)   EXACT
```

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Stefan-Boltzmann σ is a fundamental thermodynamics constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to “replace” the experimental value.

Reference

- Source: PAPER_1209EE_S624
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "stefan_sigma_5_67"})`
